Math 110 Quiz Review

Solutions

§2 Homework 8–10

Homework 8

HW8 P1.

(a) **Idea.** *Differentiation lowers degree, so enough derivatives kill any polynomial; the index is the first power that kills x^n, the slowest vector to die.*
D lowers degree by 1 and kills constants, so $D^{n+1}p = 0$ for every $p \in \mathcal{P}_n(\mathbb{C})$, while $D^n x^n = n! \neq 0$. So D is nilpotent of index $n+1$.

(b) $E(0, D) = \operatorname{null} D =$ constants, so the geometric multiplicity is 1. $G(0, D) = \mathcal{P}_n(\mathbb{C})$ since D is nilpotent, so the algebraic multiplicity is $n+1$.

(c) **Idea.** *One Jordan block means one chain. Start from the vector that takes longest to die (x^n) and differentiate repeatedly; dividing by factorials makes each step land on the next basis vector with coefficient* 1.
$f_k = x^k/k!$ for $k = 0, \ldots, n$: $Df_0 = 0$ and $Df_k = x^{k-1}/(k-1)! = f_{k-1}$. In the basis $f_0, f_1, \ldots, f_n$, column k of the matrix has a single 1 in row $k-1$: the matrix is the single block $J_{n+1}(0)$.

HW8 P2.

(a) **Idea.** *Translate the data into blocks: geometric multiplicity = number of blocks, algebraic = total size, largest block = exponent in m.*
Number of blocks = geometric multiplicity, total size = algebraic multiplicity: $J_3(2) \oplus J_1(-1) \oplus J_1(-1)$,

$$\begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}, \qquad m = (x-2)^3(x+1).$$

(b) Two blocks for 2 with total size 3: $J_2(2) \oplus J_1(2) \oplus J_1(-1) \oplus J_1(-1)$, i.e. the matrix in (a) with the $(2,3)$ entry changed to 0. The largest block for 2 has size 2, so $m = (x-2)^2(x+1)$.

HW8 P3.

(a) $N = J - I$ has $Ne_1 = 0$, $Ne_2 = e_1$, $Ne_3 = e_2$. So $\operatorname{null} N = \operatorname{span}(e_1)$; $N^2 e_3 = e_1$ and $N^2 e_1 = N^2 e_2 = 0$, so $\operatorname{null} N^2 = \operatorname{span}(e_1, e_2)$; $N^3 = 0$, so $\operatorname{null} N^3 = \mathbb{F}^3$.

(b) $G(1, J) = \operatorname{null}(J - I)^3 = \mathbb{F}^3$. Geometric multiplicity $\dim \operatorname{null}(J - I) = 1$; algebraic multiplicity $\dim G(1, J) = 3$.

(c) **Idea.** *Split J into a scalar part and a nilpotent part that commute, then use the binomial theorem; it stops at N^2 because $N^3 = 0$.*
$J = I + N$ with $IN = NI$ and $N^3 = 0$, so $J^m = I + mN + \binom{m}{2}N^2 = \begin{pmatrix} 1 & m & \binom{m}{2} \\ 0 & 1 & m \\ 0 & 0 & 1 \end{pmatrix}$ (with $\binom{m}{2} = \frac{m(m-1)}{2}$; for $m = 1$ this is J).

HW8 P4.

(a) **Idea.** *Null spaces of a block diagonal matrix add block by block, and for one block $J_j(0)^k$ is a shift by k places.* For one block $J_j(0)$, $\dim\operatorname{null} J_j(0)^k = \min(k, j)$. Add over the blocks of sizes 3 and 2: $k = 0$: 0; $k = 1$: $1+1 = 2$; $k = 2$: $2 + 2 = 4$; $k = 3$: $3 + 2 = 5$; $k = 4$: $3 + 2 = 5$.

(b) **Idea.** *Differences of the d_k count blocks of size $\geq k$; the total dimension pins down the largest size.* The number of blocks of size $\geq k$ is $\dim\operatorname{null} N^k - \dim\operatorname{null} N^{k-1}$: $2 - 0 = 2$ blocks of size ≥ 1, $4 - 2 = 2$ of size ≥ 2, $5 - 4 = 1$ of size ≥ 3. So there are two blocks, both of size at least 2, exactly one of size at least 3. The three numbers alone do not fix the larger size ($J_4(0) \oplus J_2(0)$ on $\mathbb{C}^6$ also gives $2, 4, 5$), so we use that N acts on the same 5-dimensional space as in (a): then $\operatorname{null} N^3 = V$ and $N^3 = 0$, so no block has size above 3. Hence the sizes are 3 and 2. (Equivalently: the sizes are 2 and $s \geq 3$, and $2 + s = \dim V = 5$.)

HW8 P5.

Idea. *For each eigenvalue, make the table of d_k and take differences (Prop. 77).*
Eigenvalue 1: $d_1, d_2, d_3 = 1, 2, 2$, so 1 block of size ≥ 1, 1 of size ≥ 2, none of size ≥ 3: $J_2(1)$. Eigenvalue 2: $d_1, \ldots, d_4 = 2, 3, 4, 4$, so 2 blocks of size ≥ 1, 1 of size ≥ 2, 1 of size ≥ 3, none of size ≥ 4: $J_3(2)$ and $J_1(2)$. Check: $2 + 3 + 1 = 6$. So $T \sim J_2(1) \oplus J_3(2) \oplus J_1(2)$.

HW8 P6.

(a) $\chi = (x-4)^5(x+1)$. (b) $m = (x-4)^3(x+1)$ (the largest block for 4 has size 3). (c) Eigenvalue 4: geometric 2 (two blocks), algebraic 5; eigenvalue -1: geometric 1, algebraic 1. (d) Not diagonalizable: m has the repeated root 4 (equivalently, $2 \neq 5$).

HW8 P7.

Idea. *Everything about a block diagonal matrix splits into blocks, so count block by block; for one block the null space of $J_j(0)^k$ is easy because $J_j(0)^k$ is a shift.*
Choose a basis in which N is in Jordan form with blocks $J_{j_1}(0), \ldots, J_{j_r}(0)$. N^k is block diagonal with blocks $J_{j_i}(0)^k$, so $\dim\operatorname{null} N^k = \sum_i \dim\operatorname{null} J_{j_i}(0)^k$. On one block with basis $e_1, \ldots, e_j$, $J_j(0)^k e_l = e_{l-k}$ for $l > k$ and 0 for $l \leq k$; so $\operatorname{null} J_j(0)^k = \operatorname{span}(e_1, \ldots, e_{\min(k,j)})$ and $\dim\operatorname{null} J_j(0)^k = \min(k, j)$. Then $\min(k, j) - \min(k-1, j)$ is 1 if $j \geq k$ and 0 if $j < k$. Summing over the blocks, $\dim\operatorname{null} N^k - \dim\operatorname{null} N^{k-1}$ counts the blocks with $j_i \geq k$.

Homework 9

HW9 P1.

Idea. *Trace can be computed in any basis, so use the one where P is simplest: P is 0 on $\operatorname{null} P$ and the identity on $\operatorname{range} P$.*
$V = \operatorname{null} P \oplus \operatorname{range} P$. Take a basis $u_1, \ldots, u_a$ of $\operatorname{null} P$ and $w_1, \ldots, w_b$ of $\operatorname{range} P$. $Pu_i = 0$, and $w_j = Pv$ for some v gives $Pw_j = P^2 v = Pv = w_j$. So in the basis $u_1, \ldots, u_a, w_1, \ldots, w_b$, $\mathcal{M}(P) = \operatorname{diag}(0, \ldots, 0, 1, \ldots, 1)$ with b ones, and $\operatorname{tr} P = b = \dim\operatorname{range} P = \operatorname{rank} P$.

HW9 P2.

(a) **Idea.** *m_T fixes the largest block for each eigenvalue and $\dim V = 5$ fixes the total size; list the ways to fill 5.* The largest block for 2 has size 2 and the largest block for -1 has size 1. The eigenvalues of T are exactly the roots $2, -1$ of m_T, so with $a = \dim G(2, T) \geq 2$ and $b = \dim G(-1, T) \geq 1$, Prop. 73 gives $a + b = 5$: $(a, b) \in \{(2, 3), (3, 2), (4, 1)\}$.

(b) $\operatorname{tr} T = 2a - b$: 1, 4, 7.

(c) $\operatorname{tr} = 1$: $J_2(2) \oplus J_1(-1)^{\oplus 3}$. $\operatorname{tr} = 4$: $J_2(2) \oplus J_1(2) \oplus J_1(-1)^{\oplus 2}$. $\operatorname{tr} = 7$: $J_2(2) \oplus J_2(2) \oplus J_1(-1)$ (or $J_2(2) \oplus J_1(2) \oplus J_1(2) \oplus J_1(-1)$). Each has largest block 2 for the eigenvalue 2 and only 1×1 blocks for -1, so $m_T = (x-2)^2(x+1)$.

HW9 P3.

(a) **Idea.** *Trace is the sum of the eigenvalues, and a nilpotent operator has only the eigenvalue 0.* N is nilpotent, so its only eigenvalue is 0 ($Nv = \lambda v$, $v \neq 0$ gives $0 = N^k v = \lambda^k v$). The trace is the sum of the eigenvalues with algebraic multiplicity, so $\operatorname{tr} N = 0$. (Or: in a Jordan basis $\mathcal{M}(N)$ has zero diagonal.)

(b) $\operatorname{diag}(1, -1)$: trace 0, and its square is I, so no power is 0.

(c) $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$: $AB = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $BA = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, $AB - BA = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right) \neq 0$, with trace $1 - 1 = 0$.

HW9 P4.
Idea. *Linearity and symmetry are usually immediate; test positivity by looking for a nonzero v with $\langle v, v\rangle \le 0$.*
$\langle x, y\rangle_1$: yes. It is linear in x, symmetric, and $\langle x, x\rangle_1 = x_1^2 + 2x_2^2 > 0$ for $x \neq 0$. $\langle x, y\rangle_2$: no. $\langle e_2, e_2\rangle_2 = -1 < 0$. $\langle x, y\rangle_3$: no. $\langle (1,-1), (1,-1)\rangle_3 = 0$ but $(1,-1) \neq 0$. (It is bilinear and symmetric; only positivity fails.)

HW9 P5.

(a) **Idea.** *Linearity and symmetry come from the integral; the content is positivity: a continuous function ≥ 0 with integral 0 is 0.*
Linearity in p comes from linearity of the integral; $\langle p, q\rangle = \langle q, p\rangle$; $\langle p, p\rangle = \int_{-1}^{1} p^2 \ge 0$. If $\int_{-1}^{1} p^2 = 0$, the continuous function $p^2 \ge 0$ is 0 on $[-1, 1]$, so p has infinitely many roots and $p = 0$.

(b) $\|1\|^2 = 2$, $\|x\|^2 = \int x^2 = \frac{2}{3}$, $\|x^2\|^2 = \int x^4 = \frac{2}{5}$: $\|1\| = \sqrt{2}$, $\|x\| = \sqrt{2/3}$, $\|x^2\| = \sqrt{2/5}$.

(c) $\langle 1, x\rangle = \int x = 0$ and $\langle x, x^2\rangle = \int x^3 = 0$ (odd integrands), but $\langle 1, x^2\rangle = \frac{2}{3} \neq 0$. So $1 \perp x$ and $x \perp x^2$; 1 and x^2 are not orthogonal.

HW9 P6.
Idea. *Write the left side as an inner product with the all-ones vector, then apply Cauchy–Schwarz.*
In $\mathbb{R}^3$ with the dot product, $a + b + c = \langle (a, b, c), (1, 1, 1)\rangle$, so $(a + b + c)^2 \le \|(a, b, c)\|^2 \|(1, 1, 1)\|^2 = 3(a^2 + b^2 + c^2)$. Equality holds iff (a, b, c) and $(1, 1, 1)$ are dependent, i.e. iff $a = b = c$.

HW9 P7.
Idea. *Expand both squared norms; the cross terms appear with opposite signs.*
$\|u + v\|^2 = \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle$ and $\|u - v\|^2 = \langle u, u\rangle - \langle u, v\rangle - \langle v, u\rangle + \langle v, v\rangle$. Adding, the cross terms cancel: $2\|u\|^2 + 2\|v\|^2$. This uses only (conjugate) bilinearity, so it holds over $\mathbb{R}$ and $\mathbb{C}$.

Homework 10

HW10 P1.
Idea. *Subtract shadows, then normalize. On $[-1, 1]$ odd integrands vanish, which kills many of the inner products.*
$\|1\|^2 = 2$, so $e_1 = \frac{1}{\sqrt{2}}$. $x - \langle x, e_1\rangle e_1 = x - \frac{1}{2}\int_{-1}^{1} x\,dx = x$, and $\|x\| = \sqrt{2/3}$, so $e_2 = \sqrt{\frac{3}{2}}\,x$. $x^2 - \langle x^2, e_1\rangle e_1 - \langle x^2, e_2\rangle e_2 = x^2 - \frac{1}{2}\cdot\frac{2}{3} - 0 = x^2 - \frac{1}{3}$, with $\|x^2 - \frac{1}{3}\|^2 = \int_{-1}^{1}\left(x^4 - \frac{2}{3}x^2 + \frac{1}{9}\right)dx = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45}$. So $e_3 = \sqrt{\frac{45}{8}}\left(x^2 - \frac{1}{3}\right) = \frac{\sqrt{10}}{4}(3x^2 - 1)$.
Orthonormal basis: $\frac{1}{\sqrt{2}}$, $\sqrt{\frac{3}{2}}\,x$, $\frac{\sqrt{10}}{4}(3x^2 - 1)$.

HW10 P2.
Idea. *Perpendicular to a span means perpendicular to each spanning vector: a homogeneous linear system.*
$x \in W^\perp \iff x_1 - x_2 + x_3 = 0$ and $2x_2 + x_3 = 0$. Then $x_3 = -2x_2$, $x_1 = x_2 - x_3 = 3x_2$, and x_4 is free. Basis: $(3, 1, -2, 0)$, $(0, 0, 0, 1)$. ($\dim W^\perp = 4 - 2$.)

HW10 P3.
Idea. *Same as Problem 2: impose $\langle p, 1 + x\rangle = 0$ on a general p.*
For $p = a + bx + cx^2$: $\langle p, 1 + x\rangle = \int_{-1}^{1}\left(a + (a+b)x + (b+c)x^2 + cx^3\right)dx = 2a + \frac{2}{3}(b + c)$. So $p \in W^\perp \iff 3a + b + c = 0$.
Basis: $1 - 3x$ ($a = 1, c = 0$) and $x^2 - x$ ($a = 0, c = 1$). Check: $(1 - 3x)(1 + x) = 1 - 2x - 3x^2$ integrates to $2 - 2 = 0$; $(x^2 - x)(1 + x) = x^3 - x$ integrates to 0. ($\dim W^\perp = 3 - 1 = 2$.)

HW10 P4.
Idea. *Prove the easier identity for sums, then take complements of both sides; $(Z^\perp)^\perp = Z$ undoes each $\perp$.*
First, $(X + Y)^\perp = X^\perp \cap Y^\perp$ for any subspaces: $v \perp X + Y$ iff $v \perp X$ and $v \perp Y$ (every element of $X + Y$ is $x + y$, and $X, Y \subseteq X + Y$). Apply this to $X = U^\perp$, $Y = W^\perp$ and use $(Z^\perp)^\perp = Z$ (Prop. 90): $(U^\perp + W^\perp)^\perp = U \cap W$. Taking $\perp$ of both sides and using Prop. 90 again: $U^\perp + W^\perp = (U \cap W)^\perp$.

HW10 P5.

(a) **Idea.** *The $U^\perp$ part of v is invisible to inner products with the e_j.*
Write $v = u + w$ with $u \in U$, $w \in U^\perp$ (Prop. 89); $P_U v = u$. Since $w \perp e_j$, $\langle v, e_j\rangle = \langle u, e_j\rangle$. As $e_1, \ldots, e_m$ is an orthonormal basis of U, $u = \sum_j \langle u, e_j\rangle e_j = \sum_j \langle v, e_j\rangle e_j$.

(b) **Idea.** *Split the error into a part in $U^\perp$ that every u must keep and a part in U you can remove; Pythagoras adds them in squares.*
For $u \in U$, $v - u = (v - P_U v) + (P_U v - u)$ with $v - P_U v \in U^\perp$ and $P_U v - u \in U$. By Pythagoras $\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2 \ge \|v - P_U v\|^2$. Equality iff $\|P_U v - u\| = 0$, i.e. iff $u = P_U v$.

HW10 P6.
Idea. *Riesz: find the representing vector by testing on an orthonormal basis; its i-th coordinate is $\psi(e_i)$ (no conjugates over $\mathbb{R}$).*
$\psi(f) = \int_0^2 f$ is a linear functional, so Riesz gives a unique p. Orthonormal basis of $\mathcal{P}_1(\mathbb{R})$: $e_1 = \frac{1}{\sqrt{2}}$, $e_2 = \sqrt{\frac{3}{2}}x$ (HW10 P1). $p = \psi(e_1)e_1 + \psi(e_2)e_2$ (real case, no conjugates). $\psi(e_1) = \frac{2}{\sqrt{2}} = \sqrt{2}$ and $\psi(e_2) = \sqrt{\frac{3}{2}} \cdot 2$. So $p = \sqrt{2} \cdot \frac{1}{\sqrt{2}} + 2 \cdot \frac{3}{2}x = 1 + 3x$. Check: $\langle 1, 1+3x\rangle = 2 = \int_0^2 1$ and $\langle x, 1+3x\rangle = \int_{-1}^1 3x^2 = 2 = \int_0^2 x$.

HW10 P7.
Idea. *($\Leftarrow$) If some $u \in \operatorname{range} P$ and $w \in \operatorname{null} P$ were not perpendicular, moving u slightly along w would shorten it while P still sends it to u, so P would increase a length. To make this a proof, use $\|u+tw\|^2 \geq \|u\|^2$ for all t and choose t pointing against $\langle u, w\rangle$.*
($\Rightarrow$) If $P = P_U$, then $v = P_U v + (v - P_U v)$ with orthogonal pieces, so $\|v\|^2 = \|P_U v\|^2 + \|v - P_U v\|^2 \geq \|P_U v\|^2$.
($\Leftarrow$) $V = \operatorname{null} P \oplus \operatorname{range} P$ (Prop. 91). Let $u \in \operatorname{range} P$, $w \in \operatorname{null} P$, and $t \in \mathbb{F}$. Then $P(u+tw) = u$ (as $Pu = u$, $Pw = 0$), so $\|u\|^2 \leq \|u+tw\|^2 = \|u\|^2 + 2\operatorname{Re}\big(\bar{t}\langle u, w\rangle\big) + |t|^2\|w\|^2$. Take $t = -s\langle u, w\rangle$ with $s > 0$ real: $0 \leq -2s|\langle u, w\rangle|^2 + s^2|\langle u, w\rangle|^2\|w\|^2$. Divide by s and let $s \to 0^+$: $|\langle u, w\rangle|^2 \leq 0$, so $\langle u, w\rangle = 0$. Hence $\operatorname{null} P \subseteq (\operatorname{range} P)^\perp$; both have dimension $\dim V - \dim \operatorname{range} P$, so $\operatorname{null} P = (\operatorname{range} P)^\perp$. So for $v = u + w$ with $u \in \operatorname{range} P$, $w \in \operatorname{null} P = (\operatorname{range} P)^\perp$: $Pv = Pu + Pw = u = P_{\operatorname{range} P} v$; hence $P = P_{\operatorname{range} P}$.

§3 Practice quizzes

Practice Quiz A

1. Generalized eigenspaces.

a) **Idea.** *Move powers of T between the operator and the vector: $T^{m+j+1}v = T^{m+1}(T^j v)$.*
If $T^k v = 0$ then $T^{k+1}v = T0 = 0$. Suppose $\operatorname{null} T^m = \operatorname{null} T^{m+1}$ and $v \in \operatorname{null} T^{m+j+1}$. Then $T^{m+1}(T^j v) = 0$, so $T^j v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m$, so $T^{m+j}v = 0$. Hence $\operatorname{null} T^{m+j+1} \subseteq \operatorname{null} T^{m+j}$, and with the first part they are equal.

b) **Idea.** *A strictly growing chain of dimensions in $\{0, \ldots, n\}$ can grow at most n times.*
If $\operatorname{null} T^n \neq \operatorname{null} T^{n+1}$, then by (a) no earlier pair is equal either (equality at $m \leq n$ would propagate to n). So $0 < \dim \operatorname{null} T < \cdots < \dim \operatorname{null} T^{n+1} \leq n$, which is $n+1$ distinct integers in $\{1, \ldots, n\}$, impossible. So the chain is constant from n on, and $\bigcup_k \operatorname{null} T^k = \operatorname{null} T^n$. Applied to $T - \lambda I$: $G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$.

c) **Idea.** *$E \subseteq G$; and if $T - \lambda I$ kills nothing, the chain never starts.*
If λ is an eigenvalue, $\{0\} \neq E(\lambda, T) = \operatorname{null}(T - \lambda I) \subseteq G(\lambda, T)$. If not, $\operatorname{null}(T - \lambda I) = \{0\} = \operatorname{null}(T - \lambda I)^0$, so by (a) every $\operatorname{null}(T - \lambda I)^k = \{0\}$ and $G(\lambda, T) = \{0\}$.

d) **Idea.** *Triangular, so the eigenvalues are on the diagonal. For 3, compute null spaces of powers of $A - 3I$ until two agree; for 2, the dimensions of the G's add to 3 (Prop. 73), so $G(2, A)$ is 1-dimensional and equals $E(2, A)$.*
Upper-triangular: eigenvalues $3, 3, 2$. $A - 3I = \left(\begin{smallmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{smallmatrix}\right)$ has null $= \operatorname{span}(e_1) = E(3, A)$. $(A-3I)^2 = \left(\begin{smallmatrix} 0 & 0 & -1 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ has null $= \operatorname{span}(e_1, e_2)$, and $(A-3I)^3 = \left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{smallmatrix}\right)$ has the same null space, so the chain has stopped and $G(3, A) = \operatorname{span}(e_1, e_2)$. $\dim G(2, A) = 3 - \dim G(3, A) = 1$ (Prop. 73) and $E(2, A) \subseteq G(2, A)$ is nonzero, so $G(2, A) = E(2, A) = \operatorname{null}\left(\begin{smallmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right) = \operatorname{span}\big((-1, -1, 1)\big)$. Jordan form: one block for 3 (geometric multiplicity 1) of size 2, plus $J_1(2)$: $J_2(3) \oplus J_1(2)$.

2. Trace.

a) **Idea.** *Write both traces as double sums; they differ only in the order of summation.*
$\operatorname{tr}(AB) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ji} = \sum_{j=1}^n \sum_{i=1}^m b_{ji} a_{ij} = \operatorname{tr}(BA)$.

b) **Idea.** *Group $P^{-1}AP$ as $(P^{-1}A)P$ and apply (a).*
If $B = P^{-1}AP$, then $\operatorname{tr} B = \operatorname{tr}\big((P^{-1}A)P\big) = \operatorname{tr}\big(P(P^{-1}A)\big) = \operatorname{tr} A$. Matrices of one operator in two bases are similar, so they have the same trace.

c) **Idea.** *Take traces: the left side has trace 0, the right side $\dim V > 0$. In infinite dimensions there is no trace, and the example shows $ST - TS = I$ really happens there.*
If $ST - TS = I$, then $0 = \operatorname{tr}(ST) - \operatorname{tr}(TS) = \operatorname{tr} I = \dim V > 0$, a contradiction. On $\mathcal{P}(\mathbb{R})$: $(DM - MD)p = (xp)' - xp' = p + xp' - xp' = p$, so $DM - MD = I$. The space is infinite-dimensional, so trace is not defined there and the argument does not apply.

d) **Idea.** *Translate m_T and $\dim E(1,T)$ into block sizes and counts, then fit the total dimension 4.*
The largest block for 1 has size 2, and the eigenvalue 1 has 2 blocks (geometric multiplicity 2); 3 has only 1×1 blocks, at least one. The blocks for 1 are $J_2(1), J_2(1)$ or $J_2(1), J_1(1)$; with at least one block for 3 and total size 4, only $J_2(1)\oplus J_1(1)\oplus J_1(3)$ fits. $\operatorname{tr} T = 1+1+1+3 = 6$.

Practice Quiz B

1. Counting Jordan blocks.

a) **Idea.** *N^k shifts basis vectors down by k places; count which ones fall off the end.*
With basis $e_1,\dots,e_j$, $Ne_1 = 0$ and $Ne_l = e_{l-1}$, so $N^k e_l = e_{l-k}$ for $l > k$ and $N^k e_l = 0$ for $l \le k$. The vectors e_l with $l > k$ go to distinct basis vectors, so $\operatorname{null} N^k = \operatorname{span}(e_1,\dots,e_{\min(k,j)})$ and $\dim\operatorname{null} N^k = \min(k,j)$.

b) **Idea.** *Null spaces of block diagonal matrices add; apply (a) to each block.*
N^k is block diagonal with blocks $J_{j_i}(0)^k$, so $\dim\operatorname{null} N^k = \sum_i \min(k, j_i)$ by (a). Then $\dim\operatorname{null} N^k - \dim\operatorname{null} N^{k-1} = \sum_i \big(\min(k,j_i) - \min(k-1,j_i)\big)$, and each term is 1 if $j_i \ge k$ and 0 otherwise. Exactly k: (number $\ge k$) $-$ (number $\ge k+1$) $= 2d_k - d_{k-1} - d_{k+1}$, where $d_k = \dim\operatorname{null} N^k$.

c) **Idea.** *Make the table of d_k; differences give the block counts. Then build the long chain from a vector outside $\operatorname{null}(A-2I)^2$.*
Only eigenvalue 2. $A-2I = \left(\begin{smallmatrix} 0&1&1&0\\ 0&0&1&0\\ 0&0&0&0\\ 0&0&0&0 \end{smallmatrix}\right)$ has rank 2, so $d_1 = 2$; $(A-2I)^2$ has only the $(1,3)$ entry 1, so $d_2 = 3$; $(A-2I)^3 = 0$, so $d_3 = 4$. Blocks: 2 of size ≥ 1, 1 of size ≥ 2, 1 of size ≥ 3: $J_3(2)\oplus J_1(2)$. Jordan basis: $v_3 = e_3 \notin \operatorname{null}(A-2I)^2$, $v_2 = (A-2I)e_3 = (1,1,0,0)$, $v_1 = (A-2I)v_2 = (1,0,0,0)$, and e_4 (an eigenvector not in $\operatorname{span}(v_1)$). In the basis v_1, v_2, v_3, e_4 the matrix is $J_3(2)\oplus J_1(2)$.

d) **Idea.** *Binomial theorem for the commuting pair $2I$ and N.*
$J = 2I + N$ with $N^3 = 0$ and $2I$ commuting with N: $J^k = 2^k I + k2^{k-1}N + \binom{k}{2}2^{k-2}N^2 = \left(\begin{smallmatrix} 2^k & k2^{k-1} & \binom{k}{2}2^{k-2}\\ 0 & 2^k & k2^{k-1}\\ 0 & 0 & 2^k \end{smallmatrix}\right)$.

2. Cauchy–Schwarz.

a) **Idea.** *A vector is at least as long as its shadow on v; here z is the shadow of u minus u.*
If $v = 0$, both sides are 0 and u, v are dependent. If $v \ne 0$, let $z = \frac{\langle u,v\rangle}{\|v\|^2}v - u$. Then $\langle z, v\rangle = \langle u, v\rangle - \langle u, v\rangle = 0$, so

$$0 \le \langle z, z\rangle = \langle z, -u\rangle = -\frac{\langle u,v\rangle}{\|v\|^2}\langle v, u\rangle + \|u\|^2 = \|u\|^2 - \frac{|\langle u,v\rangle|^2}{\|v\|^2}.$$

So $|\langle u,v\rangle|^2 \le \|u\|^2\|v\|^2$. Equality iff $\langle z,z\rangle = 0$ iff $z = 0$ iff u is a multiple of v; and if $u = cv$ then $|\langle u,v\rangle| = |c|\|v\|^2 = \|u\|\|v\|$.

b) **Idea.** *Compare squares and bound the cross term by Cauchy–Schwarz.*
$\|u+v\|^2 = \|u\|^2 + 2\operatorname{Re}\langle u,v\rangle + \|v\|^2 \le \|u\|^2 + 2|\langle u,v\rangle| + \|v\|^2 \le \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$.

c) **Idea.** *The parallelogram law is necessary; test it on e_1, e_2.*
A norm from an inner product satisfies the parallelogram law. With $u = (1,0)$, $v = (0,1)$: $\|u+v\|_\infty^2 + \|u-v\|_\infty^2 = 1 + 1 = 2$, but $2\|u\|_\infty^2 + 2\|v\|_\infty^2 = 4$.

d) **Idea.** *Take the inner product with the constant function 1.*
Use $\langle f, g\rangle = \int_0^1 fg$ on continuous real functions: $\left(\int_0^1 f\right)^2 = \langle f, 1\rangle^2 \le \|f\|^2\|1\|^2 = \int_0^1 f^2$. Equality iff f and 1 are dependent; since $1 \ne 0$, iff $f = c\cdot 1$ is constant.

Practice Quiz C

1. Orthonormal bases and Gram–Schmidt.

a) **Idea.** *Pythagoras, one term at a time.*
Induction on m; $m = 1$: $\|a_1e_1\|^2 = |a_1|^2$. For $m > 1$, a_me_m is orthogonal to $a_1e_1 + \cdots + a_{m-1}e_{m-1}$, so by Pythagoras $\|\sum_{i\le m} a_ie_i\|^2 = \|\sum_{i<m} a_ie_i\|^2 + |a_m|^2$. If $\sum a_ie_i = 0$, then $\sum |a_i|^2 = 0$, so every $a_i = 0$.

b) **Idea.** *Take the inner product with e_k; every other term dies.*
Write $v = \sum a_ie_i$. Then $\langle v, e_k\rangle = \sum_i a_i\langle e_i, e_k\rangle = a_k$.

c) **Idea.** *Subtract the shadows on the earlier e's; independence prevents $w_j = 0$.*
If $w_j = 0$, then $v_j \in \operatorname{span}(e_1,\dots,e_{j-1}) = \operatorname{span}(v_1,\dots,v_{j-1})$, contradicting independence. For $k < j$: $\langle w_j, e_k\rangle = \langle v_j, e_k\rangle - \sum_{l<j}\langle v_j, e_l\rangle\langle e_l, e_k\rangle = \langle v_j, e_k\rangle - \langle v_j, e_k\rangle = 0$, so $e_j \perp e_k$, and $\|e_j\| = 1$. $e_j \in \operatorname{span}(e_1,\dots,e_{j-1}, v_j) =$

$\operatorname{span}(v_1,\dots,v_j)$, so $\operatorname{span}(e_1,\dots,e_j)\subseteq\operatorname{span}(v_1,\dots,v_j)$. Both sides have dimension j (the e's are orthonormal, hence independent), so they are equal.

d) **Idea.** *Positivity: a nonzero polynomial of degree ≤ 2 cannot vanish at three points. Then run Gram–Schmidt, computing each inner product as a sum of values at $0,1,2$.*
It is bilinear and symmetric. $\langle p,p\rangle = p(0)^2+p(1)^2+p(2)^2\ge 0$, and $=0$ means p has the three roots $0,1,2$; a polynomial of degree ≤ 2 with three roots is 0. Gram–Schmidt: $\|1\|^2=3$, $e_1=\frac{1}{\sqrt{3}}$. $x-\langle x,e_1\rangle e_1 = x-\frac{0+1+2}{3} = x-1$, $\|x-1\|^2=1+0+1=2$, $e_2=\frac{x-1}{\sqrt{2}}$. $\langle x^2,e_1\rangle e_1=\frac{5}{3}$ and $\langle x^2,e_2\rangle e_2=\frac{0\cdot(-1)+1\cdot 0+4\cdot 1}{2}(x-1)=2(x-1)$, so $w_3=x^2-2x+\frac{1}{3}$, with values $\frac{1}{3},-\frac{2}{3},\frac{1}{3}$ at $0,1,2$ and $\|w_3\|^2=\frac{1}{9}+\frac{4}{9}+\frac{1}{9}=\frac{2}{3}$. $e_3=\sqrt{\frac{3}{2}}\big(x^2-2x+\frac{1}{3}\big)$.

2. Orthogonal complements and projections.

a) **Idea.** *Remove from v its shadows on an orthonormal basis of U; what is left is perpendicular to U.*
If $u\in U\cap U^\perp$, then $\langle u,u\rangle=0$ and $u=0$. Take an orthonormal basis $e_1,\dots,e_m$ of U (Gram–Schmidt). For $v\in V$, $u=\sum\langle v,e_k\rangle e_k\in U$ and $w=v-u$ has $\langle w,e_k\rangle=\langle v,e_k\rangle-\langle v,e_k\rangle=0$ for each k, so $w\perp U$. So $V=U\oplus U^\perp$, and $\dim V=\dim U+\dim U^\perp$.

b) **Idea.** *Decompose v and show the $U^\perp$ part is perpendicular to itself.*
Every $u\in U$ is orthogonal to every element of $U^\perp$, so $U\subseteq(U^\perp)^\perp$. Conversely, let $v\in(U^\perp)^\perp$ and write $v=u+w$ with $u\in U$, $w\in U^\perp$. Then $w=v-u\in(U^\perp)^\perp$ too, so $\langle w,w\rangle=0$ and $w=0$: $v=u\in U$. (Or: both have dimension $\dim V-\dim U^\perp=\dim U$.)

c) **Idea.** *Pythagoras on $v=P_Uv+(v-P_Uv)$.*
$v=P_Uv+(v-P_Uv)$ with $P_Uv\in U$ and $v-P_Uv\in U^\perp$. By Pythagoras $\|v\|^2=\|P_Uv\|^2+\|v-P_Uv\|^2\ge\|P_Uv\|^2$.

d) **Idea.** *Gram–Schmidt, then the projection formula; check that $v-P_Uv$ is perpendicular to U.*
$e_1=\frac{1}{\sqrt{2}}(1,1,0)$. $(1,0,1)-\langle(1,0,1),e_1\rangle e_1=(1,0,1)-\frac{1}{2}(1,1,0)=\frac{1}{2}(1,-1,2)$, so $e_2=\frac{1}{\sqrt{6}}(1,-1,2)$. With $v=(1,2,3)$: $\langle v,e_1\rangle e_1=\frac{3}{2}(1,1,0)$ and $\langle v,e_2\rangle e_2=\frac{5}{6}(1,-1,2)$, so $P_Uv=\big(\frac{7}{3},\frac{2}{3},\frac{5}{3}\big)$. $U^\perp$: $x+y=0$, $x+z=0$, so $U^\perp=\operatorname{span}\big((-1,1,1)\big)$. $v-P_Uv=\frac{4}{3}(-1,1,1)\in U^\perp$ (a check), and the distance is $\|v-P_Uv\|=\frac{4}{3}\sqrt{3}=\frac{4}{\sqrt{3}}$.